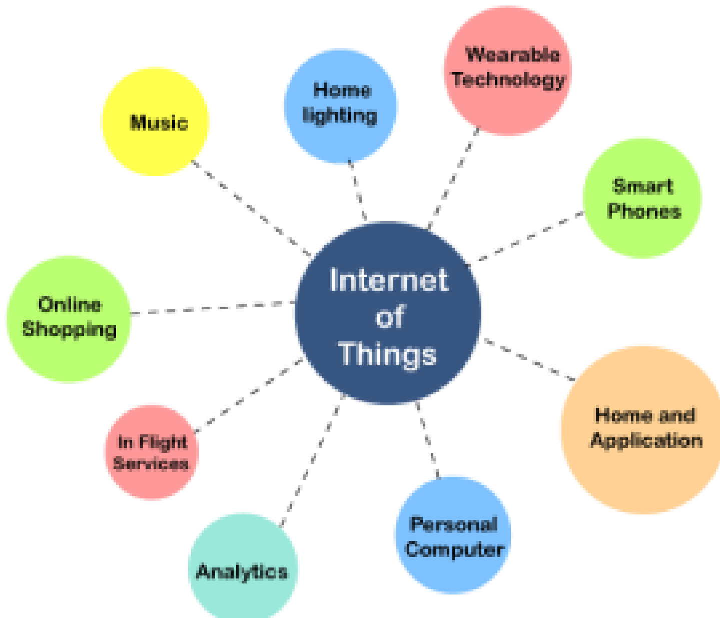
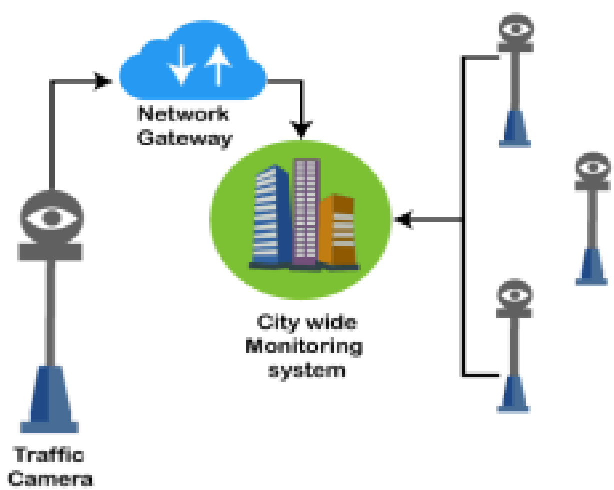
IoT applications bring a lot of value in our lives. The Internet of Things provides objects, **computing devices**, or **unique identifiers** and people's ability to transfer data across a network without the **human-to-human** or **human-to-computer interaction**.



Example 1

A traffic camera is an intelligent device. The camera monitors **traffic congestion, accidents** and **weather conditions** and can access it to a common entrance.

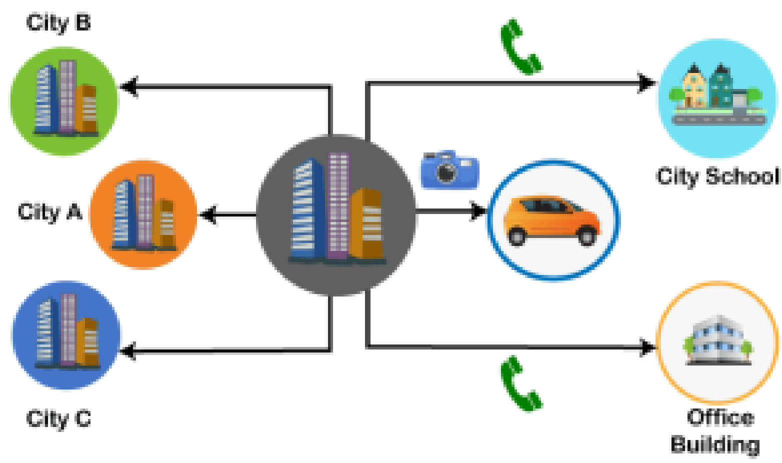
This gateway receives data from such cameras and transmits information to the city's **traffic monitoring system**.



Example 2

The municipal corporation has decided to repair a road that is connected to the national highway. It may cause traffic congestion to the national highway. The insight is sent to the traffic monitoring system.

The intelligent system analyzes the situation, estimate their impact, and relay information to other cities connected to the same highway. It generates live instructions to drivers by smart devices and radio channels.



Applications of IoT

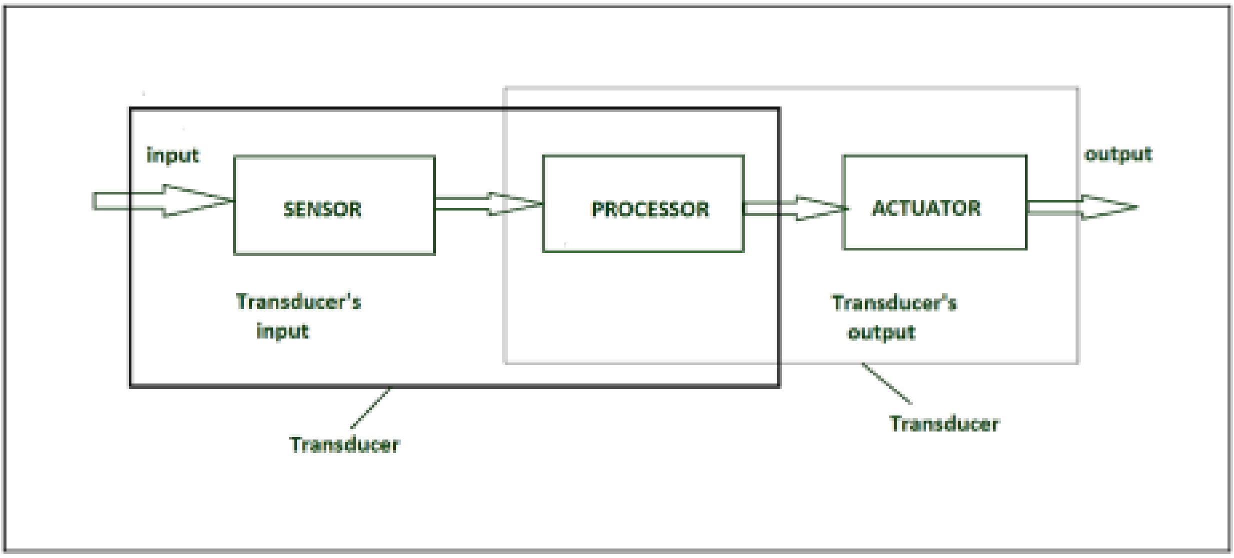
1. Wearables
2. Smart Home Applications
3. Health care
4. Smart Cities
5. Agriculture
6. Industrial Automation
7. Hacked Car
8. Smart Retail
9. Smart Supply Chain
10. Smart Farming

Sensing (Sensors in Internet of Things (IoT))

Generally, sensors are used in the architecture of IOT devices.

Sensors are used for sensing things and devices etc.
A device that provides a usable output in response to a specified measurement. The sensor attains a physical parameter and converts it into a signal suitable for processing (e.g. electrical, mechanical, optical) the characteristics of any device or material to detect the presence of a particular physical quantity. The output of

the sensor is a signal which is converted to a human-readable form like changes in characteristics, changes in resistance, capacitance, impedance, etc.



IOT HARDWARE

Transducer :

- A transducer converts a signal from one physical structure to another. • It converts one type of energy into another type.
- It might be used as actuator in various systems.

Sensor Classification :

- Passive & Active
- Analog & digital
- Scalar & vector

1. **Passive Sensor** –

   Can not independently sense the input. Ex- Accelerometer (The accelerometer works on the movement or the vibration of the body.), soil moisture (The Soil Moisture Sensor uses capacitance to measure dielectric permittivity of the surrounding medium.), water level (when the sensor is put into a certain depth in the liquid to be calculated, the pressure on the sensor's front surface is converted into the water level height) and temperature sensors (the voltage across the terminals of the diode).

2. **Active Sensor** –

   Independently sense the input. Example- Radar, sounder and laser altimeter sensors.

3. **Analog Sensor** –

The response or output of the sensor is some continuous function of its input parameter. Ex- Temperature sensor, LDR, analog pressure sensor and analog hall effect.

4. **Digital sensor –**
   Response in binary nature. Design to overcome the disadvantages of analog sensors. Along with the analog sensor, it also comprises extra electronics for bit conversion. Example – Passive infrared (PIR) sensor and digital temperature sensor(DS1620).
5. **Scalar sensor –**
   Detects the input parameter only based on its magnitude. The answer for the sensor is a function of magnitude of some input parameter. Not affected by the direction of input parameters.
   Example – temperature, gas, strain, color and smoke sensor.
6. **Vector sensor –**
   The response of the sensor depends on the magnitude of the direction and orientation of input parameter. Example – Accelerometer, gyroscope, magnetic field and motion detector sensors.

Types of sensors –

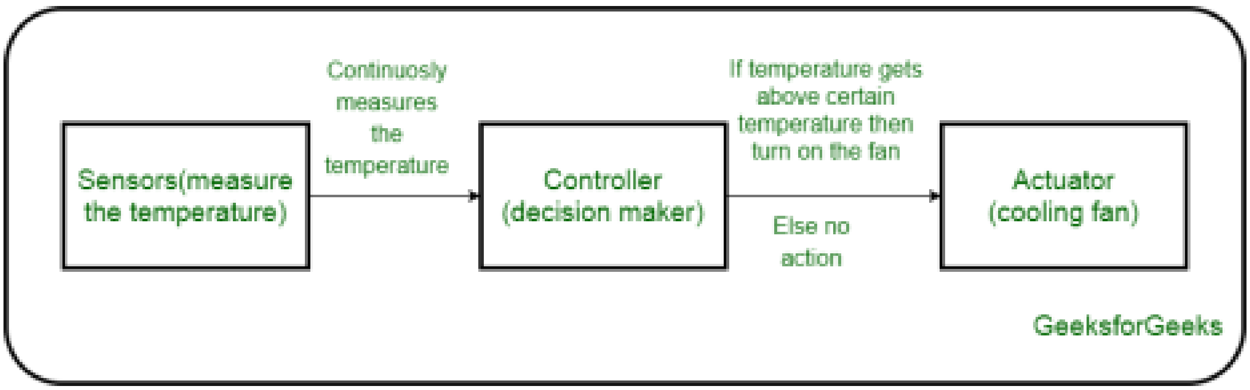
- **Electrical sensor**
- **Light sensor**
- **Touch sensor**
- **Range sensing**
- **Mechanical sensor**
- **Pneumatic sensor**
- **Optical sensor**
- **Speed Sensor**
- **Temperature Sensor**
- **PIR Sensor**
- **Ultrasonic Sensor**

Actuation (Actuators in IoT)

An actuator is a machine component or system that moves or controls the mechanism of the system. Sensors in the device sense the environment, then control signals are generated for the actuators according to the actions needed to perform.

A servo motor is an example of an actuator. They are linear or rotatory actuators, can move to a given specified angular or linear position. We can use servo motors for IoT applications and make the motor rotate to 90 degrees, 180 degrees, etc., as per our need.

The following diagram shows what actuators do; the controller directs the actuator based on the sensor data to do the work.



Working of IoT devices and use of Actuators

The control system acts upon an environment through the actuator. It requires a source of energy and a control signal. When it receives a control signal, it converts the source of energy to a mechanical operation. On this basis, on which form of energy it uses, it has different types given below.

Types of Actuators :

1. Hydraulic Actuators –

A hydraulic actuator uses hydraulic power to perform a mechanical operation. They are actuated by a cylinder or fluid motor. The mechanical motion is converted to rotary, linear, or oscillatory motion, according to the need of the IoT device. Ex- construction equipment uses hydraulic actuators because hydraulic actuators can generate a large amount of force.

Advantages :

- Hydraulic actuators can produce a large magnitude of force and high speed.
- Used in welding, clamping, etc.
- Used for lowering or raising the vehicles in car transport carriers.

Disadvantages :

- Hydraulic fluid leaks can cause efficiency loss and issues of cleaning. • It is expensive.
- It requires noise reduction equipment, heat exchangers, and high maintenance systems.

2. Pneumatic Actuators –

A pneumatic actuator uses energy formed by vacuum or compressed air at high pressure to convert into either linear or rotary motion. Example- Used in robotics, use sensors that work like human fingers by using compressed air.

Advantages :

- They are a low-cost option and are used at extreme temperatures where using air is a safer option than chemicals.
- They need low maintenance, are durable, and have a long operational life.
- It is very quick in starting and stopping the motion.

Disadvantages :

- Loss of pressure can make it less efficient.
- The air compressor should be running continuously.
- Air can be polluted, and it needs maintenance.

3. Electrical Actuators –

An electric actuator uses electrical energy, is usually actuated by a motor that converts electrical energy into mechanical torque. An example of an electric actuator is a solenoid based electric bell.

Advantages :

- It has many applications in various industries as it can automate industrial valves.
- It produces less noise and is safe to use since there are no fluid leakages.
- It can be re-programmed and it provides the highest control precision positioning.

Disadvantages :

- It is expensive.
- It depends a lot on environmental conditions.

Other actuators are –

- **Thermal/Magnetic Actuators** –

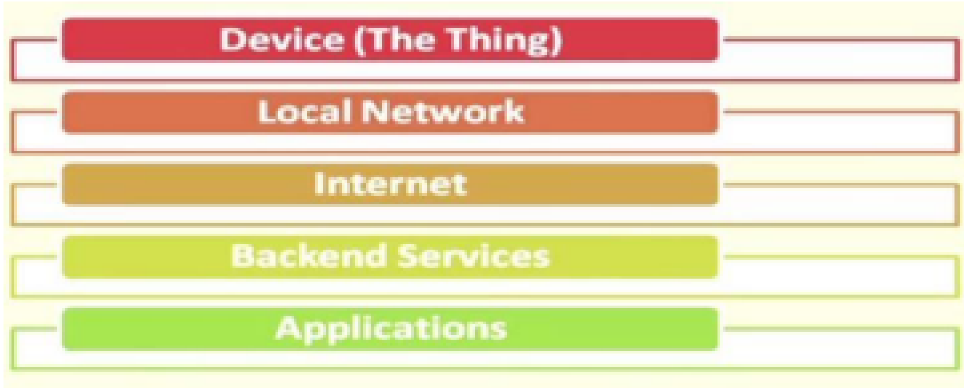
  These are actuated by thermal or mechanical energy. Shape Memory Alloys (SMAs) or Magnetic Shape-Memory Alloys (MSMAs) are used by these actuators. An example of a thermal/magnetic actuator can be a piezo motor using SMA.

- **Mechanical Actuators –**
  A mechanical actuator executes movement by converting rotary motion into linear motion. It involves pulleys, chains, gears, rails, and other devices to operate. Example – A crankshaft.
- Soft Actuators
- Shape Memory Polymers
- Light Activated Polymers
- With the expanding world of IoT, sensors and actuators will find more usage in commercial and domestic applications along with the pre-existing use in industry.

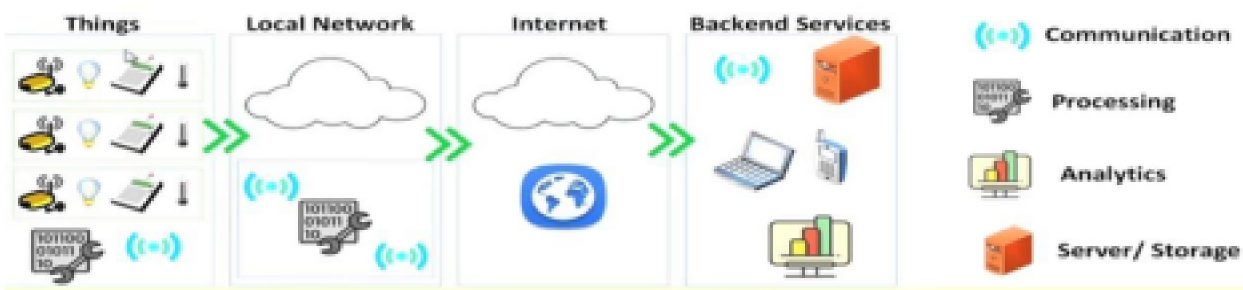
Basic of IoT Networking

Convergence of Domains

IoT Components



Example



Functional Components of IoT

• Component for interaction and communication with other IoT devices • Component for processing and analysis of operations

• Component for Internet interaction

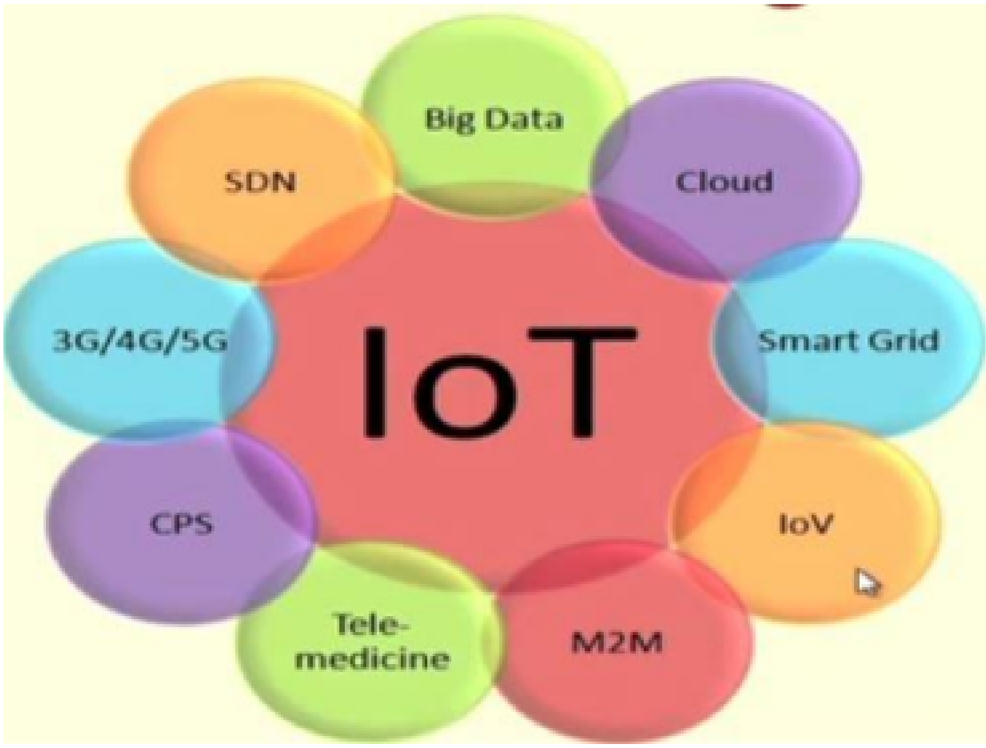
• Component for handling Web services of applications

• Component to integrate application services

• User interface to access IoT

IoT Gateways

IoT and Associated Technologies



Complexity of Networks

- Growth of networks
- Interference among devices
- Network management
- Heterogeneity in networks
- Protocol standardization within networks

Wireless Networks

- Traffic and load management
- Variations in wireless networks - Wireless Body Area
- Networks and other Personal Area Networks
- Interoperability
- Network management
- Overlay networks

Scalability

- Flexibility within Internet
- IoT integration
- Large deployment
- Real-time connectivity of billions of devices

M2M (Machine to Machine Communications)

• Machine to Machine communications, often termed M2M/IoT is going to be the next generation of Internet revolution connecting more and more devices on Internet. • M2M communications refer to automated applications which involve machines or devices communicating through a network without human intervention.

• Sensors and communication modules are embedded within M2M devices, enabling data to be transmitted from one device to another device through wired and wireless communications networks.

• M2M is expected to revolutionize the performance of various sectors, businesses and services, by providing automation and intelligence to the end devices, in a way that was never imagined before.

• It may be applied to robots and conveyor belts on the factory floor, to tractors and irrigation on the farm, from heavy equipment to hand drills, from jet engines to bus fleets; from home appliances to health monitoring; from Smart Grid to Smart Water; every piece of equipment, everywhere.

• It can bring substantial tangible social and economic benefits by giving more efficient and effective services to the citizens.

Difference between IoT and M2M

Features IoT M2M		
Abbreviation	IoT stands for the Internet of Things.	M2M stands for Machine-to Machine communication.
Intelligence	Devices include objects that are responsible for decision-making	In M2M, there is a limited amount of intelligence

	processes.	observed.
Communication Protocol Used	IoT has used internet protocols like FTP, Telnet, and HTTP.	Communication technology and Traditional protocols are uses in M2M technology.
Connection Type Used	The connection of IoT is through the network and using various types of communication.	M2M uses a point to point connection.
Scope	It has a wide range of devices, yet the scope is large.	It has a limited Scope for devices.
Business Type Used	It has Business to Consumer (B2C) and Business to Business (B2B).	It has Business to Business (B2B) communication.
Data Sharing	In IoT, data sharing depends on the Internet protocol network.	In M2M, devices may be connected through mobile or any other network.
Open API Support	IoT technology supports Open API integrations.	In M2M technology, there is no Open API support.
Example	Big Data, Cloud, Smart wearables, and many more.	Data and Information, sensors, and many more.

IoT Technology Fundamentals-Devices and gateways

IoT Technology Fundamentals-Devices

Classification of devices in IoT system is:

- Basic devices;
- Advanced devices.

Basic devices, microcontroller-class devices are those which can perform simple operations and usually cannot communicate with each other without gateways.

Advanced devices or general purpose-class devices are performing application level logic and support communication protocols.

Basic and advanced IoT devices are different in terms of scenarios and processes. Basic IoT devices are good for performing simple processes like alarms, metering, standalone smart thermostats and others.

Advanced IoT devices are designed for more complex processes, representing automated units, and processing multiple different actions at the same time.

Basic IoT devises usually serve a single operation – measure temperature, wind force or others. The hardware requirements are relatively low for this type of devices. The components used for this type of devices are inexpensive, usually using an SoC inside, containing at least two microprocessors, and several wireless connectivity stacks.

The type of communication maybe different as there is a few of them available in the market. In terms of energy performance, these devices should perform a minimum consumption. Microcontrollers are equipped with several GPIO ports for interaction with external devices, supporting I2C, and ADCs, SPI protocols as a minimum. STMicroelectronics MCU structure is depicted in Figure 1 below.

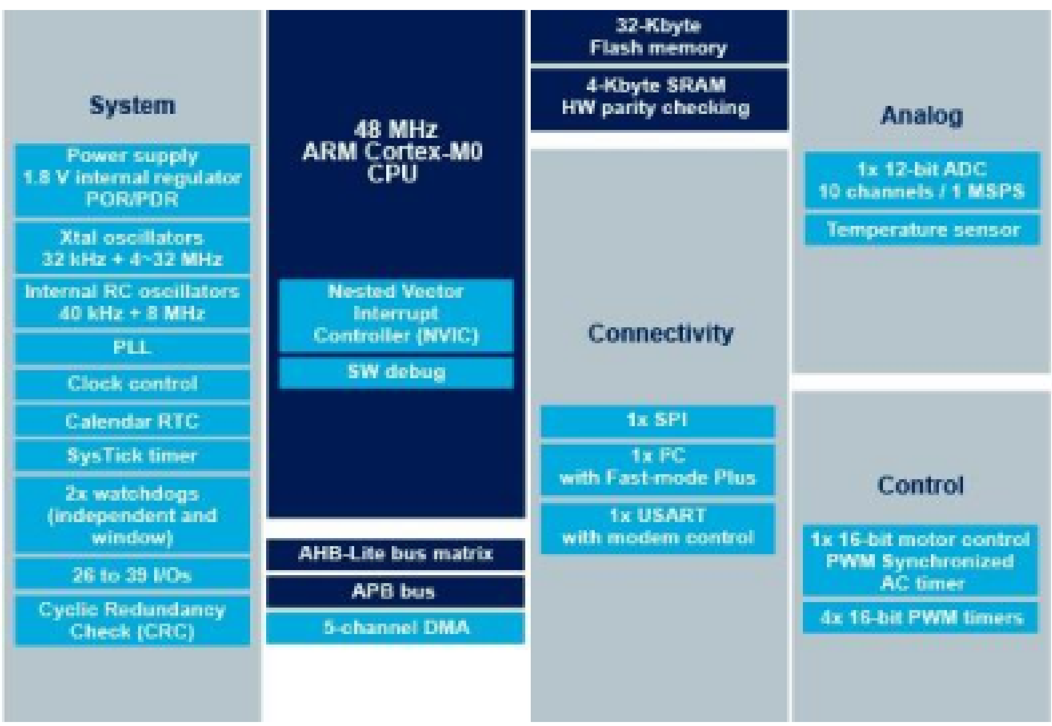


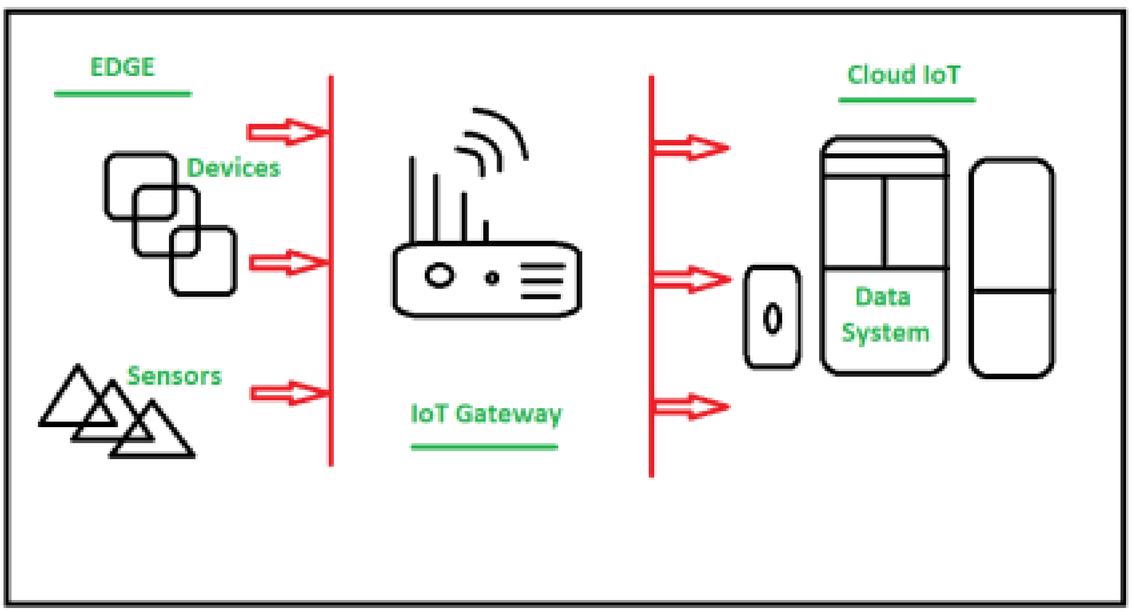
Figure 1. STMicroelectronics STM32F030C6 diagram.

UART, SPI, I2C or other serial interfaces are used to connect storage or displays, or communicate with another microcontroller. RFIC and SoC can also contain hardware security processors to provide encrypted secure communication between processors.

Wireless communication occurs by means of RF circuits. The gateway is important in this system as it provides WAN functions. Basic devices usually use the OS of FreeRTOS that support enough functionality efficiency to run simple processes.

Internet of Things (IoT) Gateways

Gateway provides a bridge between different communication technologies which means we can say that a Gateway acts as a medium to open up connections between the cloud and controller(sensors/devices) in Internet of Things (IoT). With the help of gateways, it is possible to establish device-to device or device-to-cloud communication. A gateway can be a typical hardware device or software program. It enables a connection between the sensor network and the Internet along with enabling IoT communication, it also performs many other tasks such as this IoT gateway performs protocol translation, aggregating all data, local processing, and filtering of data before sending it to the cloud, locally storing data and autonomously controlling devices based on some inputted data, providing additional device security. The below figure shows how IoT Gateways establish communication between sensors and the cloud (Data System):



As IoT devices work with low power consumption(Battery power) in other words they are energy constrained so if they will directly communicate to cloud/internet it won't be effective in terms of power. So they communicate with Gateway first using short range wireless transmission modes/network like ZigBee, Bluetooth, etc as they consume less power or they can also be connected using long range like Cellular and WiFi etc. Then Gateway links them to Internet/ cloud by converting data into a standard protocol like MQTT.
using ethernet, WiFi/cellular or satellite connection. And in mostly Gateway is

Mains powered unlike sensor nodes which are battery powered. In practice there are multiple Gateway devices. Let's think about a simple IoT gateway, then our smartphone comes into picture as it can also work as a basic IoT gateway when we use multiple radio technologies like WiFi, Bluetooth, Cellular network of smart phone to work on any IoT project in sending and receiving data at that time this also acts as a basic IoT Gateway.

Key functionalities of IoT Gateway :

- Establishing communication bridge
- Provides additional security.
- Performs data aggregation.
- Pre processing and filtering of data.
- Provides local storage as a cache/ buffer.
- Data computing at edge level.
- Ability to manage entire device.
- Device diagnostics.
- Adding more functional capability.
- Verifying protocols.

Working of IoT Gateway :

1. Receives data from sensor network.
2. Performs Pre processing, filtering and cleaning on unfiltered data.
3. Transports into standard protocols for communication.
4. Sends data to cloud.

Advantages of Gateway:

There are several advantages of using a gateway in the Internet of Things (IoT), including:

- **Protocol translation:** IoT devices typically use different communication protocols, and a gateway can translate between these protocols to enable communication between different types of devices.
- **Data aggregation:** A gateway can collect data from multiple IoT devices and aggregate it into a single stream for easier analysis and management.
- **Edge computing:** Gateways can perform edge computing tasks such as data processing, analytics, and machine learning, enabling faster and more efficient decision-making.
- **Security**: Gateways can act as a secure access point for IoT devices, providing a layer of protection against cyber threats.

- **Scalability:** Gateways can support a large number of IoT devices and can be easily scaled up or down to meet changing needs.
- **Improved reliability:** Gateways can help to improve the reliability of IoT devices by managing network connectivity and providing a backup mechanism in case of network failure.
- **Cost-effective**: Gateways can be a cost-effective way to manage and control a large number of IoT devices, reducing the need for expensive infrastructure and IT resources.

Data Management Systems for The IoT Devices

The Internet of Things (IoT) device management enables users to track, monitor and manage the devices to ensure these work properly and securely after deployment.

Billions of sensors interact with people, homes, cities, farms, factories, workplaces, vehicles, wearables and medical devices, and beyond. The Internet of Things (IoT) is changing our lives from managing home appliances to vehicles. Devices can now advise us about what to do, when to do and where to go. Industrial applications of the IoT assist us in managing processes, and predicting faults and disasters. The IoT platforms help set and maintain parameters to refine and store data accordingly.

Steps in IoT data management

Data management is the process of taking the overall available data and refining it down to important information. Different devices from different applications send large volumes and varieties of information. Managing all this IoT data means developing and executing architectures, policies, practices and procedures that can meet the full data lifecycle needs.

Things are controlled by smart devices to automate tasks, so we can save our time. Intelligent things can collect, transmit and understand information, but a tool will be required to aggregate data and draw out inferences, trends and patterns.

The IoT data management

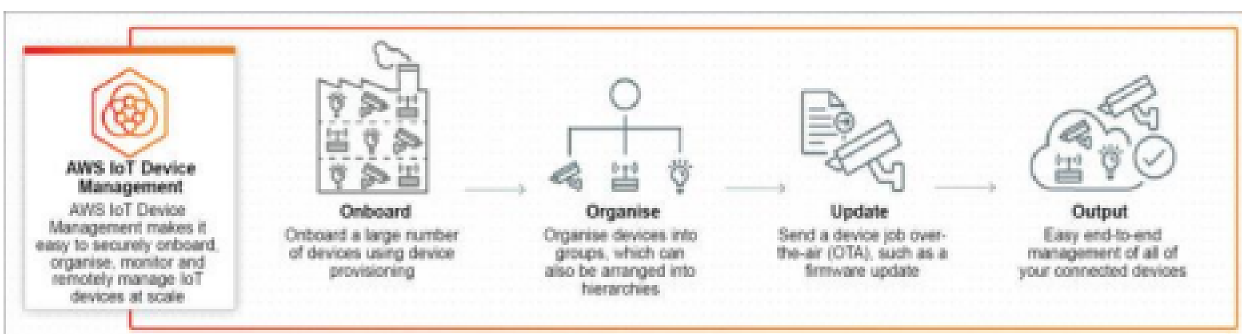
In edge computing, data is processed near the data source or at the edge of the network. While in a typical cloud environment, data processing happens in a centralised data storage location. By processing and using some data locally, the IoT saves storage space for data, processes information faster and meets security challenges.

Edge computing, data governance policies and metadata management help firms deal with issues of scalability and agility, security and usability. This further assist them decide whether to manage data on the edge or only after sending it to the cloud.

Sensors produce a large amount of data for edge gateway devices so that these can make decisions by analysing the data. These high-performance systems not only need to collect data in real time but also to organise and provide data to other systems.

AWS IoT device management

This is critical across industrial, consumer and commercial applications such as an industrial and connected home. It enables customers to manage large and diverse device fleets such as operational technology systems, cameras, machines, vehicles, appliances and more.



Working of AWS IoT device management

It enables monitoring of usage and performance metrics for devices across industrial sectors such as manufacturing, oil and gas, and mining. It allows monitoring metadata and policy changes with service alerts to inform about any adjustments to the devices' configuration. It also allows detection of any unusual behaviour across a device and to take mitigating actions.

It provides secure onboarding, organising, monitoring, troubleshooting and sending of firmware updates over-the-air (OTA). It enables adding device attributes like device name, type and manufacturing year, certificates and access policies to the IoT Registry. Then, it assigns them to devices, and makes connected devices ready for service quickly. This helps to quickly search and find any IoT device across the entire device fleet in near real time.

It is easy to find devices based on a combination of attributes like device ID, device state and type, so that action can be taken, in addition to troubleshooting. Actions such as reboots, factory resets and security patches can also be remotely executed.

How is IoT transforming businesses?

In the Business Enterprise, IoT not only means to connect the devices to the Internet, however, it is more than that. Now, IoT is transforming the business enterprises by creating the opportunities to get smarter about the product, services, and the customer experience.

There are several ways through IoT is transforming the business. Some of them are mentioned below:

- Improving Customer Experience
- Greater Efficiency
- More Data - More Opportunity
- Creating New Business Models
- Cost Reduction and Gain Productivity
- Asset Tracking and Waste Reduction

Improving Customer Experience

Due to growth in technology, IoT is playing an opportunity to both customers as well as service provider by understanding the customer behavior and their requirement. When product and service provider understand how its customers use their product they can better fulfill their needs and improve customer experience. As IoT data offer the real time operation, companies can respond quickly to issues and request as they arise.

More Data - More Opportunity

IoT provides power to business enterprises to collect and analyze more data from various sources such as from their local setups and networks. This data lead more opportunity for automatic product updates or upgrades, tracing and tracking the assets. As the large volume of data flow from data centers, production systems, sensors and IoT systems to the business enterprise at real or non-real-time helps the enterprise to offer opportunities for innovation and growth.

Greater Efficiency

Business enterprise is also looking towards more flexibility and efficiency. The higher efficiency will be achieved through IoT as it expands over various technologies and parts of the organization. The capabilities of product or services are realized as data streaming from sensors is analyzed, and changes are deployed without human intervention.

Creating New Business Models

The IoT also allows a business enterprise to transform their conventional models into new revenue streams. This IoT data can also be shared across an enterprise's ecosystem of partners and customers which provides new paths to innovation in the form of new value-added services and continuous engagement.

Cost Reduction and Gain Productivity

As the IoT connects devices and keeps the business key's process data over the networks, leads the smart inventory management, waste management and the cost reduction. Due to IoT, leaders can easily identify the ways to boost efficiency and productivity to enhance potential revenue stream.

Role of Cloud Computing in IoT

- Cloud computing is helping the IoT in getting success. Cloud is a big factor in the success of IoT.
- As cloud enable user to carry and access all thing over internet without any storage, IoT is related with cloud computing.
- Future users of these technologies will gain a number of benefits. • As was already mentioned, cloud computing allows for scalability in the delivery of applications and software as a service by enabling businesses to manage and store data across cloud platforms.

Why Cloud Computing Is Essential for IoT?

- The cloud is an excellent IoT enabler that satisfies the data-driven requirements of the company.
- Cloud also offers technology framework. Using that framework, we can develop better IoT devices.

- Speed and scale are two essential aspects of cloud computing, and they work in unmatched harmony with IoT networking and mobility.
- So, user can benefit more by combine use of cloud computing and IoT.

Some factors show that the cloud is necessary for the success of IoT, and here are some of them.

Cloud Functions as a Distant Computing Power

On-premises infrastructure reliance is no longer a viable option. As usage of cloud and IoT devices are increasing day by day, we generated large amount of data. We need to process that data quickly using big data. The advantage of having a tonne of storage capacity in this situation comes from the cloud. As we are heading towards 5G from 4G, cloud computing also enables developer more speed in getting access to data.

IoT Data is More Secure And Private as a Result of Cloud Computing.

IoT involves significant data generation. And when you work with data, the data security and data privacy become issue. IoT also makes use of mobility. Cloud uses advance encryption algorithms and authentication. Which enable cloud to provide its user high security.

No Requirement For Hosting on-premises

For IoT devices, plug-and-play hosting services are necessary. This will become quite expensive due to plug-and-play hosting services. This will cost more to organizations. This type of hosting services needs hardware system. Due to the combined power of cloud computing and IoT, you do not need to depend on substantial machinery. As cloud computing infrastructure make it ready to use without having hardware storage device set-up offline. This makes it easy for IoT hosting organizations.

Improved Device-To-Device Communication

We can use cloud technology for the communication using the IoT. Smart device can easily connect with each other using IoT APIs. It also makes internal communication between devices fast and easy.

Less Cost of Ownership

While preventing enterprises from putting up the infrastructure, cloud technology also provides many resources. As a result, it saves lot of money on infrastructure construction. Additionally, because there is no idea of local systems, hardware, and software in the cloud, the IT teams are abler to concentrate on their regular tasks.

Program For Business Continuity

Business continuity is guaranteed by cloud computing, even if unexpected disasters occur while it is being used. There is no danger of data loss because data is maintained on additional distinct servers, which is considerably more crucial in the case of IoT-based architecture.

IoT innovations with low entry barriers require hassle-free hosting options. As a result, cloud computing in IoT is a suitable solution. IoT players can use the power of distant data centers due to cloud computing without requiring on premises gear and software. IoT cloud computing is the best option financially because users need to adhere to the pay-as-you-go concept. It also saves a tonne of money upfront.

This helps businesses can launch massive IoT projects with ease. This removes many obstacles to entry for the majority of IoT-based organizations.

Communication Between Devices

By using cloud computing in proper way, IoT devices can communicate with each other seamlessly. As a result, connected devices and smart devices can communicate with various reliable APIs. In this way, networked technologies are made possible by cloud computing.